

Linear Momentum Balance (2D) in Cartesian coordinate

For x-component

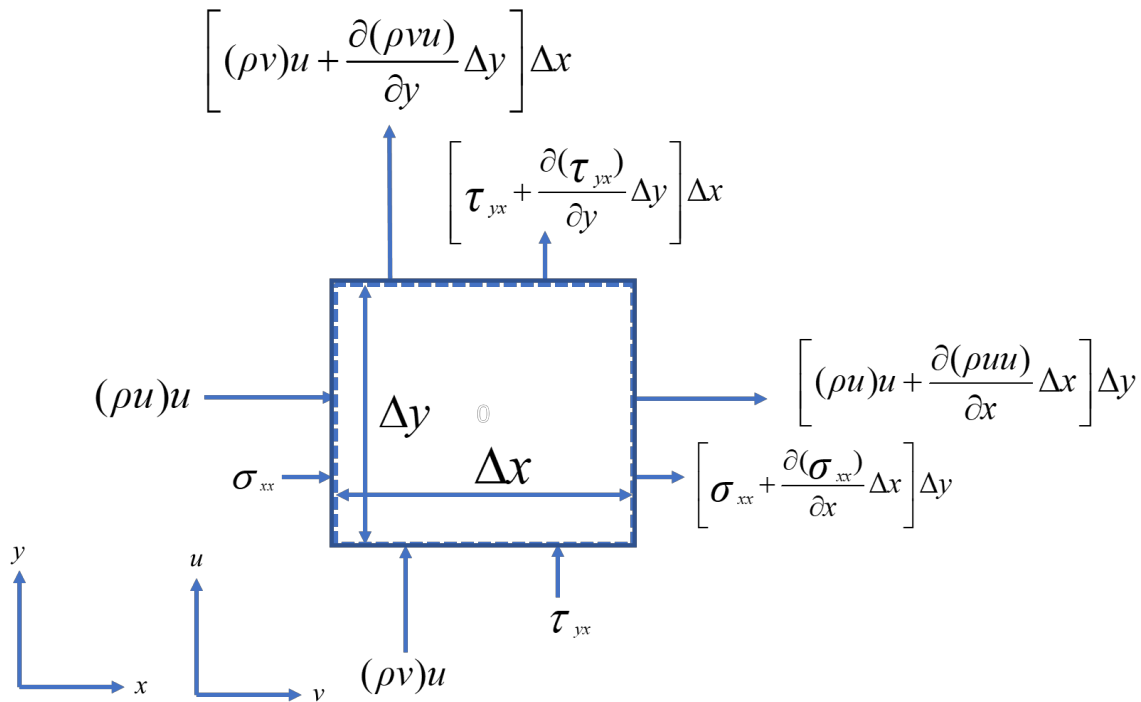


Figure 1. x- Component of the Linear momentum balance in 2D flow field (Cartesian coordinate system)

τ_{yx} = Viscous flux of x momentum in y direction

In general (for special surfaces where the direction of normal to the surface is z, y, or z directions)

τ_{ij} = 'i' is the direction normal to the surface along which the force is acting and 'j' is the direction of the applied force.

For any arbitrary surface, applied notation T_i^n means 'n' is the direction normal to the surface and 'i' is the direction of applied force.

A general expression for the linear momentum balance:

$$[\text{Rate of momentum accumulation}]_1 = [\text{Rate of momentum in}]_2 - [\text{Rate of momentum out}]_3 + [\text{Sum of the forces acting on the system}]_4 \quad (1)$$

Momentum transfer to or from the system takes place by 2 mechanisms

i) Convection (by the virtue of bulk fluid flow)

ii) Molecular transfer/Diffusion

1) Rate of momentum accumulation (in 'x' direction)

$$\frac{\partial(\rho u \Delta x \Delta y)}{\partial t}$$

2) Rate of momentum in

$$\Delta y[(\rho u)u] + \Delta y[\sigma_{xx}] + \Delta x[(\rho u)v] + \Delta x[\tau_{yx}]$$

3) Rate of momentum out

$$[(\rho u)u + \frac{\partial(\rho uu)}{\partial x} \Delta x] \Delta y + [\sigma_{xx} + \frac{\partial}{\partial x}(\sigma_{xx}) \Delta x] \Delta y + [(\rho u)v + \frac{\partial(\rho uv)}{\partial y} \Delta y] \Delta x + [\tau_{yx} + \frac{\partial}{\partial y}(\tau_{yx}) \Delta y] \Delta x$$

4) Sum of all forces acting

a) Arising from fluid pressure P

$$\text{in} = P_x \Delta y$$

$$\text{out} = [P_x + \frac{\partial P_x}{\partial x} \Delta x] \Delta y$$

b) Body force

$$[\text{let 'X' be the B.F per unit volume}] = X \Delta x \Delta y$$

Substituting into equation (1)

$$\begin{aligned} \frac{\partial(\rho u \Delta x \Delta y)}{\partial t} = & \Delta y[(\rho u)u] + \Delta y[\sigma_{xx}] + \Delta x[(\rho u)v] + \Delta x[\tau_{yx}] - \Delta y(\rho u)u - \frac{\partial(\rho uu)}{\partial x} \Delta x \Delta y \\ & - \Delta y \sigma_{xx} - \frac{\partial}{\partial x}(\sigma_{xx}) \Delta x \Delta y - \Delta x(\rho u)v - \frac{\partial(\rho uv)}{\partial y} \Delta y \Delta x - \Delta x \tau_{yx} - \frac{\partial}{\partial y}(\tau_{yx}) \Delta y \Delta x \\ & + P_x \Delta y - \Delta y P_x - \frac{\partial P_x}{\partial x} \Delta x \Delta y + X \Delta x \Delta y \end{aligned}$$

After simplification,

$$\frac{\partial(\rho u)}{\partial t} = -\frac{\partial(\rho uu)}{\partial x} - \frac{\partial \sigma_{xx}}{\partial x} - \frac{\partial(\rho uv)}{\partial y} - \frac{\partial \tau_{yx}}{\partial y} - \frac{\partial P_x}{\partial x} + X$$

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho uu)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} = -\frac{\partial \sigma_{xx}}{\partial x} - \frac{\partial \tau_{yx}}{\partial y} - \frac{\partial P_x}{\partial x} + X \quad (2)$$

$$\text{LHS} = \rho \frac{\partial u}{\partial t} + u \frac{\partial \rho}{\partial t} + \rho u \frac{\partial u}{\partial x} + u \frac{\partial(\rho u)}{\partial x} + \rho u \frac{\partial v}{\partial y} + v \frac{\partial(\rho u)}{\partial y}$$

$$= \rho \frac{\partial u}{\partial t} + u \frac{\partial \rho}{\partial t} + u \left[u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} \right] + \rho u \frac{\partial u}{\partial x} + \rho u \frac{\partial v}{\partial y} + v \left[u \frac{\partial \rho}{\partial y} + \rho \frac{\partial u}{\partial y} \right]$$

$$= \rho \left[\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] + u \left[\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + v \frac{\partial \rho}{\partial y} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \right]$$

$$= \rho \frac{Du}{Dt} + u \left[\frac{D\rho}{Dt} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \right] = \rho \frac{Du}{Dt}$$

$$[\text{As, } \frac{D\rho}{Dt} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0, \text{ from continuity equation}]$$

Then,

$$\rho \frac{Du}{Dt} = - \left[\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} \right] - \frac{\partial P}{\partial x} + X \quad (3)$$

Equation 3 is the general linear momentum balance (2D) equation in the x-direction of the cartesian coordinate system.

Substitution of the relevant expression of σ_{xx} and τ_{yx} using constitutive relations for Newtonian fluids will lead to the famous Navier Stokes Equations

Energy Transport

A general expression for energy transport

$$\begin{aligned}
 [\text{Rate accumulation in the CV}]_1 = & [\text{Net transfer of energy by fluid flow}]_2 + \\
 & [\text{Net heat transfer by conduction}]_3 + \\
 & [\text{Rate of internal heat generation}]_4 - \\
 & [\text{Net work transfer from the CV to its environment}]_5 \quad (1)
 \end{aligned}$$

For terms **1** and **2**,

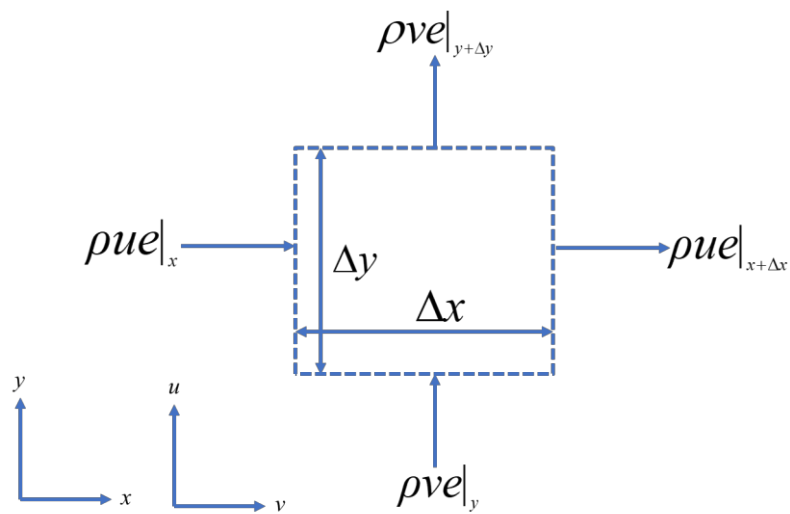


Figure 2. Energy transport by fluid flow in a 2D flow field (Cartesian coordinate system)

Define e as the total specific energy of the system,

$$e = \hat{u} + \frac{1}{2}v^2$$

$$\hat{u} = \text{specific internal energy} = U/m = \hat{u}$$

$$\frac{1}{2}v^2 = \text{specific Kinetic energy} = KE/m = \frac{1}{2} \frac{mv^2}{m} = \frac{1}{2}v^2$$

1. Rate of energy accumulation

$$\frac{\partial(\rho e)}{\partial t} \Delta x \Delta y \quad (2)$$

2. Net transfer of energy by fluid flow

i) Rate of energy in

$$\Delta y \rho u e + \Delta x \rho v e$$

ii) Rate of energy out

$$\Delta y \rho u e|_{x+\Delta x} + \Delta x \rho v e|_{y+\Delta y} = [\rho u e + \frac{\partial(\rho u e)}{\partial x} \Delta x] \Delta y + [\rho v e + \frac{\partial(\rho v e)}{\partial y} \Delta y] \Delta x$$

$$\text{Net} = (\text{in} - \text{out}) = (i - ii)$$

$$= -\left[\frac{\partial(\rho u e)}{\partial x} + \frac{\partial(\rho v e)}{\partial y}\right] \Delta x \Delta y \quad (3)$$

3. Net heat transfer by conduction

Conduction heat flux is the transport element,

$$q'' = \frac{\text{Conductive heat}}{\text{Area} * \text{time}}$$

Similarly, Net conduction heat transfer,

$$-\left[\frac{\partial(q''_x)}{\partial x} + \frac{\partial(q''_y)}{\partial y}\right] \Delta x \Delta y \quad (4)$$

4. Rate of Internal heat generation

Let the volumetric heat generation rate be q''' (heat/volume*time)

Thus, the total internal heat generation is,

$$q''' \Delta x \Delta y \quad (5)$$

5. Net work done by the fluid element against its surrounding

It consists of two parts

- i) *Work against the body forces (originate from volume)*
- ii) *Work against the surface forces*
 - a) Work against the pressure forces
 - b) Work against the viscous forces

Recall,

Work = (force) $\times$ (distance in the direction of the applied force)

Rate of work = (force) $\times$ (velocity in the direction of the applied force)

- i) *Work against the body forces (originate from volume)*

$$-\Delta x \Delta y \rho [v_x g_x + v_y g_y] \quad 6(i)$$

g_x, g_y are the gravitational accelerations in x and y directions respectively ~ Force/mass.

The negative sign appears in Eqn.5(i) as displacement is against the body force.

- ii) *Work against the surface forces*

- a) Pressure

Net force (similar to 2 and 3)

$$-\left[\frac{\partial (pv_x)}{\partial x} + \frac{\partial (pv_y)}{\partial y} \right] \Delta x \Delta y \quad 6 \text{ [ii(a)]}$$

b) Viscous forces

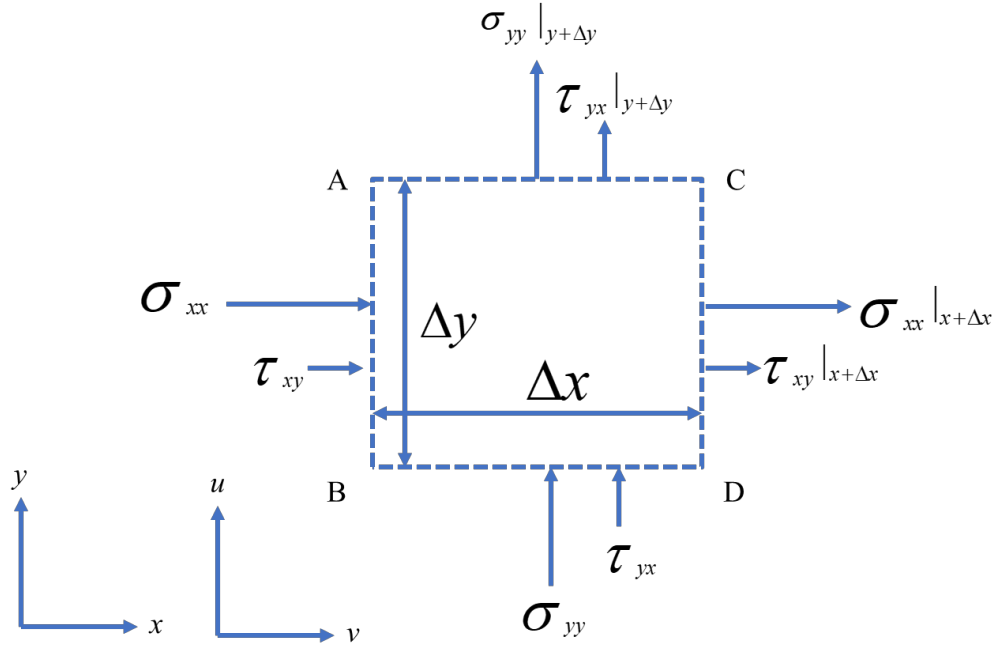


Figure 4. Work done against the viscous force in a 2D flow field (Cartesian coordinate system)

The arrow directions for the τ_{xy} and τ_{yx} are not physical.

The general representation is τ_{ij} , where i is the direction normal to the surface along which force is acting, and j is the direction of the applied force.

For the AB plane (normal to the x -axis at $x=0$), forces/unit area acting are σ_{xx} and τ_{xy} .

The velocity in the direction of the force σ_{xx} will be u (Since the direction of applied force, the j^{th} index, is x)

The velocity in the direction of the force τ_{xy} will be v (Since the direction of applied force, the j^{th} index, is y).

Hence, the rate of work done by the viscous forces at the AB plane against the surrounding (sign not considered) is

$$[\sigma_{xx}u + \tau_{xy}v]\Delta y$$

Similarly, at the BD plane (perpendicular to the y -axis at $y=0$)

Work done by the viscous force

$$[\sigma_{yy}v + \tau_{yx}u]\Delta x.$$

Thus, net work done by the viscous force-

$$\begin{aligned}
 &= [\sigma_{xx}u + \tau_{xy}v] \Big|_x \Delta y + [\sigma_{yy}v + \tau_{yx}u] \Big|_y \Delta x - [\sigma_{xx}u + \tau_{xy}v] \Big|_{x+\Delta x} \Delta y - [\sigma_{yy}v + \tau_{yx}u] \Big|_{y+\Delta y} \Delta x \\
 &= -\left[\frac{\partial \sigma_{xx}u}{\partial x} + \frac{\partial \tau_{xy}v}{\partial x} + \frac{\partial \sigma_{yy}v}{\partial y} + \frac{\partial \tau_{yx}u}{\partial y} \right] \Delta x \Delta y \\
 &= -\left[u \frac{\partial \sigma_{xx}}{\partial x} + \sigma_{xx} \frac{\partial u}{\partial x} + \tau_{xy} \frac{\partial v}{\partial x} + v \frac{\partial \tau_{xy}}{\partial x} + v \frac{\partial \sigma_{yy}}{\partial y} + \sigma_{yy} \frac{\partial v}{\partial y} + u \frac{\partial \tau_{yx}}{\partial y} + \tau_{yx} \frac{\partial u}{\partial y} \right] \Delta x \Delta y
 \end{aligned}$$

Writing the constitutive equations,

$$\begin{aligned}
 \tau_{yx} &= \tau_{xy} = -\mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\
 \sigma_{xx} &= -2\mu \frac{\partial u}{\partial x} + \frac{2}{3} \mu (\nabla \cdot v) \\
 \sigma_{yy} &= -2\mu \frac{\partial v}{\partial y} + \frac{2}{3} \mu (\nabla \cdot v)
 \end{aligned}$$

For incompressible flow, $\nabla \cdot v = 0$

Substituting τ_{yx} , σ_{xx} , σ_{yy} with $\nabla \cdot v = 0$ in the above equation we get

$$[\mu \phi] \Delta x \Delta y \quad \quad \quad 6 \text{ [ii(b)]}$$

Where,

$$\phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2 = \text{Viscous dissipation function}$$

Net work done by fluid on the surroundings,

$$5(i) + 5 \text{ [ii(a)]} + 5 \text{ [ii(b)]}$$

$$= -[\rho(ug_x + vg_y)]\phi + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} - \mu\phi] \Delta x \Delta y \quad (6)$$

Substituting (2), (3), (4), (5) and (6) into equation (1)

$$\frac{\partial(\rho e)}{\partial t} = -\left[\frac{\partial(\rho ue)}{\partial x} + \frac{\partial(\rho ve)}{\partial y} \right] - \left[\frac{\partial(q_x'')}{\partial x} + \frac{\partial(q_y'')}{\partial y} \right] + q''' - [\rho(ug_x + vg_y)]\phi + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} - \mu\phi]$$

$$\rho \frac{\partial(e)}{\partial t} + e \left[\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} \right] + \rho u \frac{\partial e}{\partial x} + \rho v \frac{\partial e}{\partial y} = -(\nabla \cdot \vec{q}) + \dot{q} + \rho(\vec{v} \cdot \vec{g}) - \nabla \cdot (\rho \vec{v}) + \mu \phi$$

$$\rho \frac{\partial(e)}{\partial t} + e \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) \right] = -(\nabla \cdot \vec{q}) + \dot{q} + \rho(\vec{v} \cdot \vec{g}) - \nabla \cdot (\rho \vec{v}) + \mu \phi$$

$$\text{Putting } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \text{ (continuity),}$$

$$\rho \frac{\partial(e)}{\partial t} = -(\nabla \cdot \vec{q}) + \dot{q} + \rho(\vec{v} \cdot \vec{g}) - \nabla \cdot (\rho \vec{v}) + \mu \phi \quad (7)$$

Now, gravitational force per unit mass $\vec{g}$ is a vector and can be expressed as a gradient of a scalar,

$$\vec{g} = -\nabla \phi \quad \text{Where, } \phi \text{ is the potential energy}$$

Hence,

$$\rho(\vec{v} \cdot \vec{g}) = -\rho(\vec{v} \cdot \nabla \phi) = -\rho \frac{D\phi}{Dt} + \rho \frac{\partial \phi}{\partial t}$$

Also,

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \nabla \cdot (v\phi) \approx \frac{\partial \phi}{\partial t} + \vec{v} \cdot \nabla \phi;$$

$$(\vec{v} \cdot \nabla \phi) = \frac{D\phi}{Dt} - \frac{\partial \phi}{\partial t};$$

If ϕ is time-independent then, $\frac{\partial \phi}{\partial t} = 0$

$$\text{So, } \rho(\vec{v} \cdot \vec{g}) = -\rho \frac{D\phi}{Dt}$$

Substituting into equation 7,

$$\rho \frac{D(e + \phi)}{Dt} = \dot{q} - \nabla \cdot (P\vec{v}) + \mu \phi - \nabla \cdot \vec{q} \quad (8)$$

This is an equation of change for total energy, $e + \phi = \hat{u} + \frac{v^2}{2} + \phi$

Now we have to use the laws of thermodynamics to get the transport equation in terms of Temperature.

According to the First law of thermodynamics

$$Q = \Delta u + w$$

So, the energy supplied to the system is used to increase the internal energy only. Considering internal energy only, equation 8 can be rewritten as

$$\rho \frac{De}{Dt} = q''' - \nabla \cdot (P\vec{v}) + \mu\phi - \nabla \cdot q'' \quad (9)$$

Where $e = \hat{u}$ only.

Using thermodynamic relation Enthalpy of the system,

$$H = U + PV$$

In terms of specific enthalpy,

$$H / m = h = (U / m + PV / m) = [\hat{u} + P / (m / V)] = [e + P / \rho]$$

Or,

$$\frac{Dh}{Dt} = \frac{De}{Dt} + \frac{1}{\rho} \frac{DP}{Dt} - \frac{P}{\rho^2} \frac{D\rho}{Dt} \quad (10)$$

Using Fourier's law –

$$q'' = -K\nabla T \quad (11)$$

Substituting equation 10 and 11 in equation 9, we get

$$\begin{aligned} \rho \frac{Dh}{Dt} &= q''' - P(\nabla \cdot \vec{v}) + \mu\phi + \frac{DP}{Dt} - \frac{P}{\rho} \frac{D\rho}{Dt} - \nabla \cdot q'' \\ \rho \frac{Dh}{Dt} &= q''' + \mu\phi + \frac{DP}{Dt} - \frac{P}{\rho} \left[\frac{D\rho}{Dt} + \rho(\nabla \cdot \vec{v}) \right] + \nabla \cdot (K\nabla T) \\ \rho \frac{Dh}{Dt} &= q''' + \mu\phi + \frac{DP}{Dt} + \nabla \cdot (K\nabla T) \end{aligned} \quad (12)$$

Now, $dh = C_p dT$ is only valid for ideal gases (see **Table 1**)

In general, we know,

$$dH = TdS + VdP$$

[As $H = U + PV$ and, $dH = (dU + PdV) + VdP = dQ + VdP = TdS + VdP$]

Table 1: Summary of thermodynamic relations and models

	Internal Energy $du = T ds - P dv$	Enthalpy $dh = T ds + v dP$	Entropy $ds = \frac{1}{T} du + \frac{P}{T} dv$
Pure substance	$du = c_v dT$ $+ \left[T \left(\frac{\partial P}{\partial T} \right)_v - P \right] dv$	$dh = c_p dT$ $+ \left[-T \left(\frac{\partial v}{\partial T} \right)_p + v \right] dP$	$ds = \frac{c_p}{T} dT - \left(\frac{\partial v}{\partial T} \right)_p dP$ $= \frac{c_v}{T} dT + \left(\frac{\partial P}{\partial T} \right)_v dv$
Ideal gas	$du = c_v dT$	$dh = c_p dT$	$ds = c_p \frac{dT}{T} - R \frac{dP}{P}$ $= c_v \frac{dT}{T} + R \frac{dv}{v}$ $= c_v \frac{dP}{P} + c_p \frac{dv}{v}$
Incompressible liquid	$du = c dT$	$dh = c dT + v dP$	$ds = c \frac{dT}{T}$

In terms of specific property

$$dh = T ds + \frac{1}{\rho} dP \quad (13)$$

Now, $s = f(T, P)$

$$ds = \left(\frac{\partial s}{\partial T} \right)_P dT + \left(\frac{\partial s}{\partial P} \right)_T dP \quad (14)$$

From Maxwell's relation $\left[\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P \right]$,

we have (for specific properties)

$$\left(\frac{\partial s}{\partial P} \right)_T = - \left(\frac{\partial (1/\rho)}{\partial T} \right)_P = \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_P = \frac{-\beta}{\rho}$$

Where is β the thermal expansion coefficient

$$\beta = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_P = \frac{1}{(1/\rho)} \left(\frac{\partial (1/\rho)}{\partial T} \right)_P = - \frac{\rho}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_P = - \frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P$$

We also have, $\left(\frac{\partial s}{\partial T} \right)_P = \frac{C_P}{T}$

Substituting into equation 14, we have

$$ds = \frac{C_p}{T} dT - \frac{\beta}{\rho} dP$$

Substituting into equation 13, we have

$$dh = T \left[\frac{C_p}{T} dT - \frac{\beta}{\rho} dP \right] + \frac{1}{\rho} dP = C_p dT + \frac{1}{\rho} (1 - \beta T) dP$$

Taking substantial derivative and multiplying by ρ we get,

$$\rho \frac{Dh}{Dt} = \rho C_p \frac{DT}{Dt} + (1 - \beta T) \frac{DP}{Dt} \quad (15)$$

(Relation between h and T)

Using equation 15 and 12, we get the **general form of the energy transport equation for Newtonian fluids**

$$\rho C_p \frac{DT}{Dt} = q''' + \mu \phi + \beta T \frac{DP}{Dt} + \nabla \cdot (K \nabla T) \quad (16)$$

(Common form of $\mu \phi$ is $-\nabla \cdot (\tau \cdot \vec{v})$)

Special cases,

(A) For ideal gas, $\beta = 1/T$

So,

$$\rho C_p \frac{DT}{Dt} = q''' + \mu \phi + \frac{DP}{Dt} + \nabla \cdot (K \nabla T) \quad (17)$$

(B) For incompressible liquid, $\beta = 0$

$$\rho C_p \frac{DT}{Dt} = q''' + \mu \phi + \nabla \cdot (K \nabla T) \quad (18)$$

In most of the convection problems we have

- 1) Constant K
- 2) Zero heat generation
- 3) Negligible viscous dissipation, $\phi = 0$

Using the above three assumptions, the temperature distribution model will be

$$\rho C \frac{DT}{Dt} = K \nabla^2 T \quad (19)$$

For an incompressible fluid, (specific heat, $C = C_p$)